CLINICAL · EAR

The deaf child and congenital hearing loss

Speech and language are built between birth and five years, and they are built on hearing. A child who does not speak has usually not heard. Children identified and helped before six months of age do far better, which is why every newborn is screened, and why a failed screen is never brushed aside.

AT A GLANCE

- Identify and treat before speech develops.
- Screen all newborns: half have no risk factor.
- OAE and ABR, then age-appropriate testing.
- Aids early; implant for severe-profound loss.
- Check ECG, thyroid, kidney and eyes by syndrome.

1 CHAPTER ONE Overview

The critical years.

1.1 Why early matters

The auditory pathways develop in response to sound during the first years of life. Without input, the auditory cortex is taken over by other functions and language develops late or not at all. Early fitting of hearing aids or a cochlear implant allows near-normal language in many children.

1.2 Prelingual and postlingual

Prelingual deafness occurs before speech has developed and has the greatest effect on language; *postlingual* deafness occurs after.

1.3 The 1-3-6 rule

Screen by one month, diagnose by three months, and start intervention by six

months.

2 CHAPTER TWO Causes

Genes, infection and the newborn period.

2.1 Proportions

About half of congenital sensorineural loss is genetic, about a quarter environmental and about a quarter unexplained. Of the genetic cases, about 70% are non-syndromic (mostly autosomal recessive) and 30% syndromic.

2.2 Genetic

2.2.1 Non-syndromic

GJB2 mutations affecting connexin 26 (DFNB1) account for up to half of recessive non-syndromic deafness. Mitochondrial mutations (12S rRNA, 1555A>G) cause deafness after even a single dose of an aminoglycoside.

2.2.2 Syndromic

SYNDROMES TO KNOW

SYNDROME	CLUE	HEARING LOSS
Waardenburg	White forelock, odd-coloured eyes, dystopia canthorum (type 1)	Sensorineural
Usher	Retinitis pigmentosa	Sensorineural
Pendred	Goitre, enlarged vestibular aqueduct	Sensorineural
Jervell and Lange-Nielsen	Fainting, long QT	Sensorineural
Alport	Nephritis, X-linked	Progressive sensorineural
Treacher Collins	Lower lid coloboma, small jaw and malar, microtia	Conductive

SYNDROME	CLUE	HEARING LOSS
Crouzon and Apert	Craniosynostosis, proptosis (syndactyly in Apert)	Conductive or mixed
Branchio-oto-renal	Neck fistulae, pits, renal anomalies	Conductive or mixed
Stickler	Small jaw, cleft, myopia	Conductive or sensorineural
Goldenhar	Facial asymmetry, epibulbar dermoid	Conductive or mixed
Klippel-Feil	Short neck, fused cervical spine	Sensorineural or mixed
Down	Effusion, narrow canals	Conductive



FIGURE 1 Waardenburg syndrome: white forelock, heterochromia of the irides and depigmented skin.
Dhingra, fig. 19.1

SYNDROME PEARLS

- Usher type 1: profound loss and absent vestibular function; type 2: moderate loss, normal or reduced vestibular function.
- Pendred: the goitre appears around age eight; the perchlorate discharge test is positive.
- Waardenburg type 1 has dystopia canthorum; type 2 does not.

- Treacher Collins: lower lid coloboma; Goldenhar: upper lid.

2.3 Inner ear malformations

- **Michel:** complete absence of the inner ear; no aid or implant can help.
- **Mondini:** a cochlea of only one and a half turns, often with an enlarged aqueduct.
- **Scheibe:** cochleosaccular dysplasia, the commonest.
- **Alexander:** the basal turn alone, losing high tones.
- **Bing-Siebenmann:** the membranous labyrinth fails to develop.
- **Enlarged vestibular aqueduct:** fluctuating, progressive loss after minor head injury.

WHERE THE BOOKS DIFFER · ENLARGED VESTIBULAR AQUEDUCT

Dhingra	Lee
Wider than 2 mm.	1.5 mm or more.

Avoid contact sports and head injury; surgical closure of the aqueduct causes hearing loss.

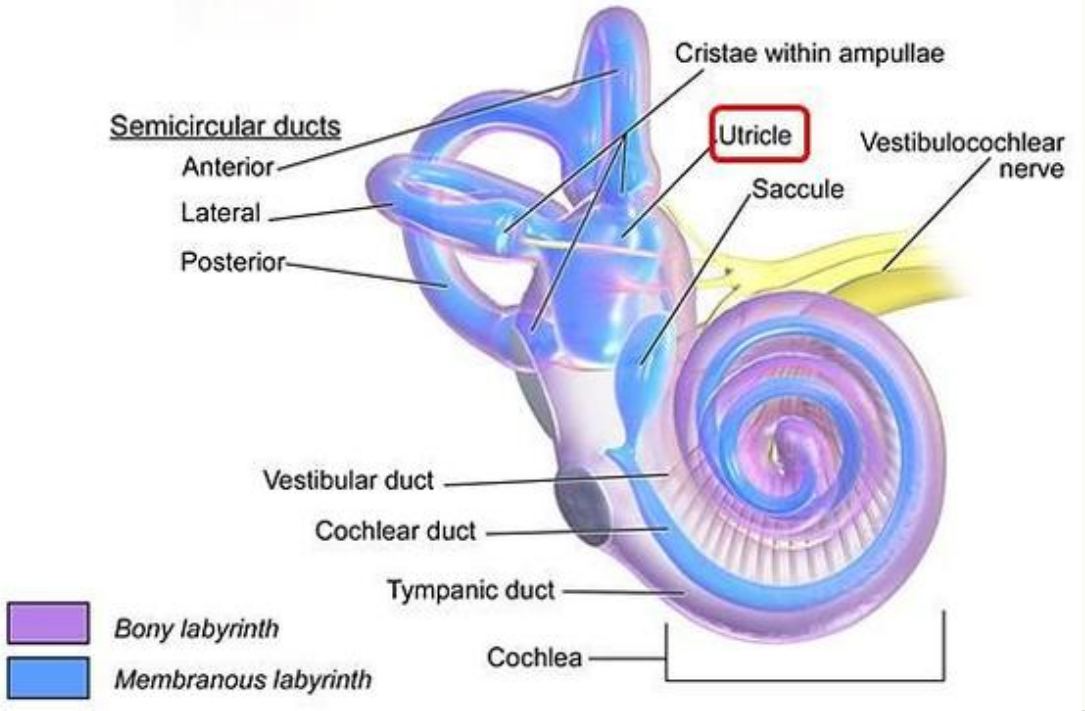


FIGURE 2 The membranous labyrinth within the bony labyrinth, the structure that fails to form in these malformations.

Anatomy atlas · ENT Steps p. 72

2.4 Environmental

ENVIRONMENTAL CAUSES

- PRENATAL **TORCH infections** cytomegalovirus, rubella, toxoplasmosis, syphilis, Zika; aminoglycosides, quinine, thalidomide, alcohol, radiation
- PERINATAL **Hypoxia, prematurity** birth weight under 1500 g, jaundice (bilirubin over 20 mg%), sepsis, NICU stay, ventilation
- POSTNATAL **Meningitis** viral infections, otitis media with effusion, ototoxic drugs, trauma, noise

2.4.1 Cytomegalovirus

The commonest non-genetic cause. The loss can be absent at birth and appear later, so a normal newborn screen does not exclude it.

2.4.2 Meningitis

Labyrinthitis ossificans can close the cochlea within weeks; test hearing and image quickly.

2.5 Risk factors

Family history, craniofacial anomaly or syndrome stigmata, NICU stay of five days or more, ventilation, low Apgar scores, severe jaundice, congenital infection, and meningitis. Half of children with sensorineural loss have no risk factor at all, which is why screening must be universal.

3 CHAPTER THREE Evaluation

Screen, confirm, find the cause.

3.1 Newborn screening

3.1.1 Otoacoustic emissions

Test the outer hair cells; quick and cheap. Absent with fluid or debris, or a cochlear loss. Normal in auditory neuropathy, which they therefore miss.

3.1.2 Automated ABR

Tests the pathway to the brainstem; used for NICU babies, who are at risk of auditory neuropathy.

3.2 Diagnostic audiology

3.2.1 ABR

Ear-specific thresholds, under natural sleep or sedation.

3.2.2 Behavioural tests by age

- Under six months: behavioural observation.
- Six to eighteen months: distraction testing.
- Six months to two years: visual reinforcement audiometry.
- Two to five years: play audiometry.
- Older: conventional and speech audiometry.

3.3 Finding the cause

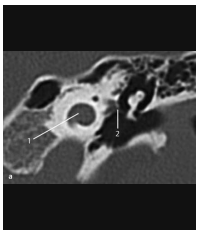
Imaging (CT and MRI) before implantation; ECG for long QT; urinalysis; thyroid function; eye examination; CMV PCR on newborn saliva, urine or dried blood spot (within 21 days); and genetic testing (GJB2, GJB6 and gene panels).

LONG QT

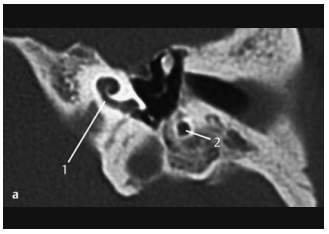
A deaf child who faints needs an urgent ECG: Jervell and Lange-Nielsen syndrome can cause sudden death. Treat with beta-blockers.

IMAGING IN THE APP 5 cases

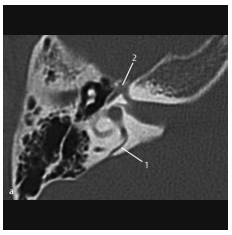
Inner ear and ossicular malformations, an enlarged vestibular aqueduct, Treacher Collins syndrome, and a gusher ear.



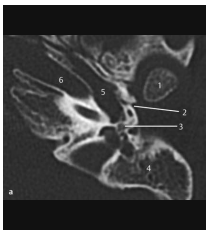
Malformations of the Inner Ear
Congenital · 3 images



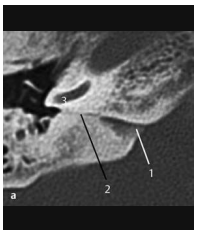
Malformations of the Inner Ear and Ossicular Chain
Congenital · 3 images



Vestibular Aqueduct, Enlargements
Congenital · 11 images



Malformations, Treacher Collins
Congenital · 2 images



Gusher
Infection & inflammation · 3 images

3.4 Signs at home

No startle to loud sound, sleeping through noise, no babbling or no words by one to two years, and poor school performance.

4 CHAPTER FOUR Management

Hearing first, then language.

4.1 Counselling

Parents need a clear explanation, support and early contact with educators.

4.2 Hearing aids

As early as possible, in both ears if there is usable hearing.

4.3 Cochlear implant

For severe to profound loss with little benefit from aids, ideally around the first birthday, or urgently after meningitis.

OPERATIVE GUIDES IN THEATRE 5 operations

Pediatric Cochlear Implantation
Paediatric implant
Congenital Malformations of the Middle Ear
Congenital middle ear malformations
Congenital and Acquired Atresia and Stenosis of the External Auditory Canal
Atresia
Microtia
Microtia
Bone-Anchored Hearing Devices
Bone-anchored devices

4.4 Communication and education

Auditory-oral approaches, sign language, or total communication; radio aids at school; speech and language therapy; vocational guidance later.

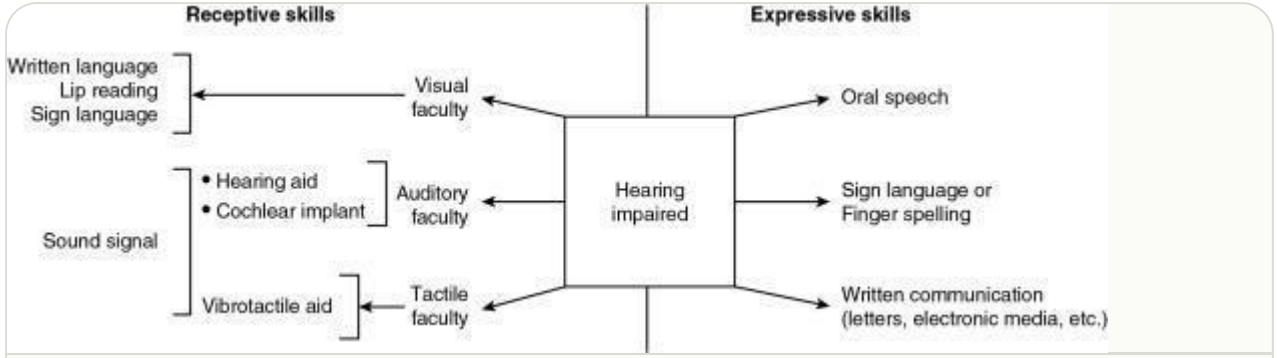


FIGURE 3 The faculties a hearing-impaired child can use for receptive and expressive communication.
Dhingra, fig. 19.3

4.5 Specific treatments

- **Symptomatic congenital CMV:** valganciclovir for six months.
- **Toxoplasmosis:** pyrimethamine and sulfadiazine can prevent hearing loss.
- **Long QT:** beta-blockers.
- **Pendred goitre:** thyroid hormone.
- **Enlarged aqueduct:** avoid head injury and contact sport.

4.6 Follow-up

Children with risk factors need continued hearing checks even after passing the newborn screen; treated CMV infants continue audiology until about age six.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 On call

A FAILED NEWBORN HEARING SCREEN

1. Repeat the screen; if it fails again, arrange a diagnostic ABR. Don't reassure.
2. A deaf child with fainting: urgent ECG for long QT.
3. After meningitis: audiology and imaging quickly, before ossification closes the implant window.

5.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes:** genetic, prenatal, perinatal, postnatal.
- **High-risk registry.**
- **OAE versus ABR,** and auditory neuropathy.
- **Behavioural tests by age.**
- **The syndromes table.**
- **Inner ear malformations,** especially Michel and Mondini.

5.3 Common pitfalls

COMMON PITFALLS

- Reassuring parents after a failed screen.
- Relying on OAE in a NICU baby.
- Missing late-onset CMV loss after a normal newborn screen.
- Forgetting the ECG.

Written for this app from Oxford Textbook of Otolaryngology ch. 16, 83, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 1, 19 and K. J. Lee ch. 14, 17. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · CONGENITAL

Malformations of the Inner Ear

IMAGING FINDINGS

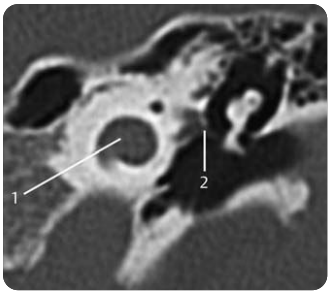
- Abnormal cochlea, vestibule or canals on CT and high-resolution T2
- IP-II: cystic apex, normal basal turn, large vestibular aqueduct
- Common cavity, cochlear hypoplasia, IP-I and IP-III by appearance

TOP DIFFERENTIALS

- Labyrinthitis ossificans
- Normal variant of canal size

CHECKLIST

- Cochlear nerve present on oblique sagittal T2
- Cochlear nerve canal width; gusher risk (IP-I, IP-III)



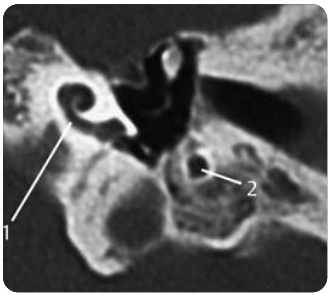
CT, coronal. This coronal view of the cochlea does not show any bony modiolus (1). This empty cochlea is suggestive of an incomplete partition…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1102.

INNER EAR & IAC · CONGENITAL

Malformations of the Inner Ear and Ossicular Chain

- This section illustrates the importance of a systematic comparison of both sides as well as the different compartments of the ear since bilateral congenital pathology may show (subtle) variations between the left and right side.



CT, axial. Illustrative evaluation of malformations of the inner ear as well as the middle ear. This axial slice demonstrates a cochlea consisting of…

INNER EAR & IAC · CONGENITAL

Vestibular Aqueduct, Enlargements

IMAGING FINDINGS

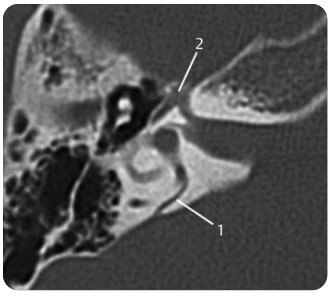
- Vestibular aqueduct wider than about 1 mm at the midpoint or 2 mm at the operculum
- Large endolymphatic sac on T2
- Often with IP-II cochlear deficiency

TOP DIFFERENTIALS

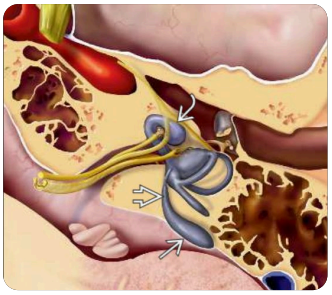
- Normal aqueduct (compare with the posterior canal)
- IP-I cystic cochleovestibular malformation

CHECKLIST

- Look at both ears: often bilateral
- Pendred syndrome association



CT, axial. The normal vestibular aqueduct (1) in its passage from the vestibule to its junction with sinuses in the region of the sigmoid sinus. Ther…



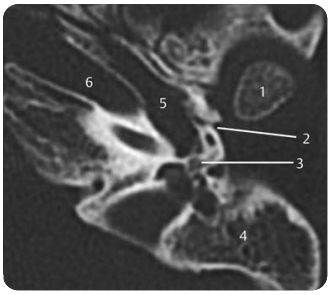
In left inner ear, large endolymphatic sac epidural and intraosseous components are shown. Cochlea is malformed, with absent septation…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104.

MIDDLE EAR & MASTOID · CONGENITAL

Malformations, Treacher Collins

- In deformities of the temporal bone, pay particular attention to the:
- External auditory canal, bony or fibrous atresia, appearance and position of the mandibular condyle and temporomandibular joint.
- Appearance of the middle ear cavity and mastoid pneumatization.
- Signs of ankylosis or disconnections of the ossicular chain.
- Presence of inner ear deformities, the round and oval windows, and the vestibular aqueduct.
- Aberrations in the course and anterior displacement of the facial nerve, which may complicate surgery.

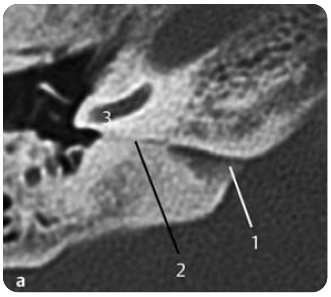


CT, axial. For orientation, this is a slice through the basal cochlear turn and the mandibular condyle (1). Behind the condyle, the bony part of the external…

INNER EAR & IAC · INFECTION & INFLAMMATION

Gusher

- A wide communication between the subarachnoid space and the perilymph (a defect at the fundus of the IAC or a dilated cochlear aqueduct) lets CSF pour out when the footplate is opened.
- CT clues: a bulbous, widened lateral IAC with a deficient fundus, absent modiolus and a corkscrew cochlea (X-linked incomplete partition type III, POU3F4).
- In a boy with a congenital mixed or conductive loss, review the CT for these signs before any stapes surgery.



CT, axial. At stapedotomy, abundant flow of inner ear fluid, the gusher phenomenon, made further surgical exploration impossible and the hole i…